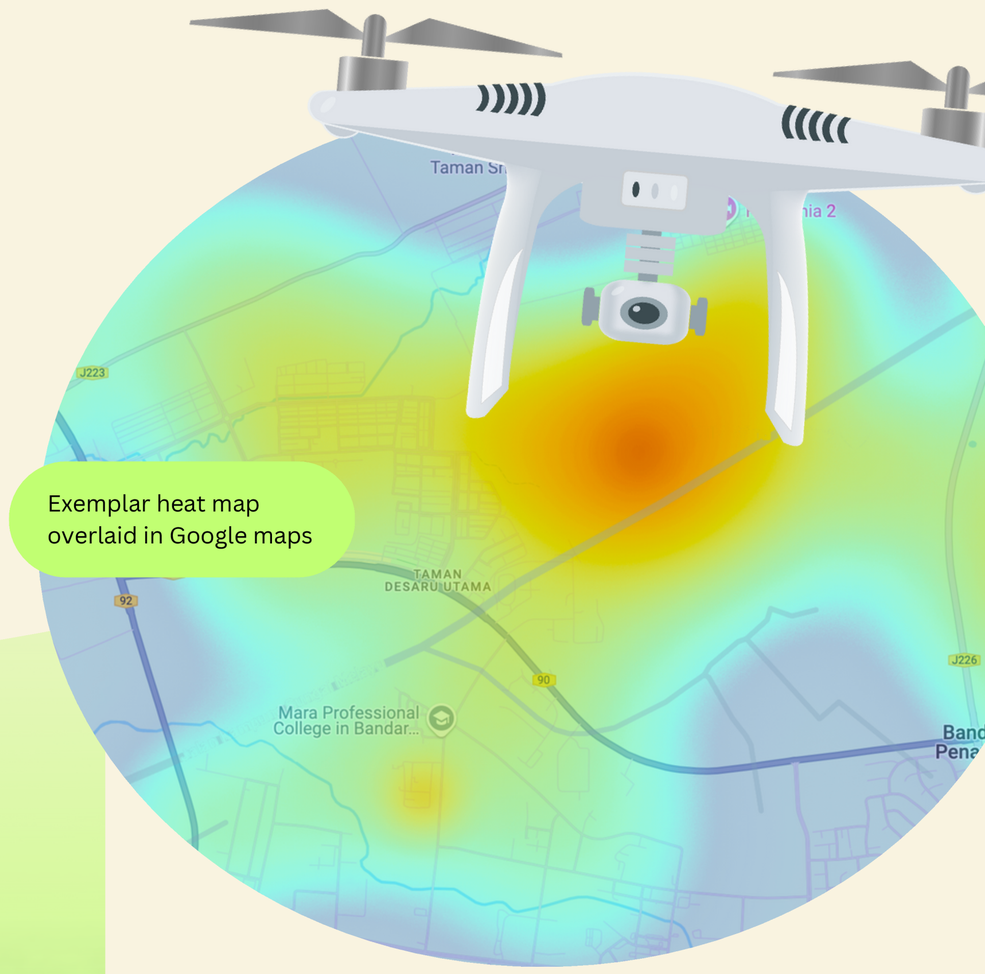
AI RECON DRONE

We have developed a “Recon Drone AI Software” powered by the YOLO-v8 AI model, capable of using drone photos to scan large areas, including forest, urban areas and beaches, to generate heat maps that highlight trash hotspots. These heat maps allow smaller teams to quickly target high density trash areas, significantly improving waste management by eliminating the need for larger teams to manually search through expansive areas.

The system is scalable and can adapt to various environments and sizes, making it suitable for both dense urban settings and expansive rural landscapes.



HOW OUR SOLUTION WORKS

Our solution works using a drone and our personally trained AI. The steps we take to convert photos into heat maps are as follows:

Using any drone, a user can fly it around an area, taking aerial photos to gather data on the environment bellow, including the location of rubbish.

Each image, along with its individual GPS data, is processed by our custom-trained AI model, which detects trash. For each image, a 2D array is generated, recording the x- and y-coordinates and the number of trash items identified.

The 2D arrays are used to create a heat map overlay on Google Maps, displaying trash density across the scanned area

Anyone in the area and cleanup teams can access the heat maps to help them conduct clean ups in areas that need more attention due to high rubbish density. This will lead to quicker and more effective cleanups, significantly improving waste management and benefiting the environment.

THE TEAM



Shi Hao Ng

Lead Machine Learning Engineer



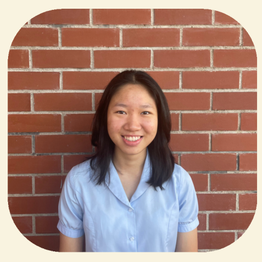
Olivier Bhogal

Creative Director and Assistant Pipeline Engineer



Louis Gan

Lead Drone Engineer and Assistant Pipeline Engineer



Jolie Teo

Lead Pipeline Engineer and Assistant Machine Learning Engineer



Jonatas Mancuso

Lead Front-end Developer

TECHNOLOGIES USED

Google Colab: Used for our initial testing and training our YOLOv8 CNN model (with Nvidia A100 PCIe GPUs). This setup allowed us to run initial, small-scale experiments

Google Cloud Compute Engine: We rented powerful Nvidia H100 GPUs for large-scale training jobs and handle extensive data processing.

Google Maps API: Based on the coordinates of each drone images, this API is then used to provide a clear visualisation of our trash detection results by overlaying heat maps in our downstream webapp to show the clean up teams where trash are.

Google Cloud AI Platform: To host our trained model and ensure it could handle multiple inference requests while scaling according to demand.

Google Firebase: To handle database management, user authentication, and data storage. This allowed for a real-time, responsive web and mobile application.

Raspberry Pi & Nvidia Jetson Nano: For real-time inference, allowing the drone to detect trash in remote areas.

DJI Drone: To capture high-resolution images to display trash detection results on frontend

Google Drive: Central storage for datasets, models, and all our project files.

Google Docs: To write detailed documentation and collaborate on project write-ups.

Google Sheets: To track data, manage annotations, and keep track of our project budget.

Google Meet: Facilitated our virtual meetings, enabling us to brainstorm and resolve challenges efficiently.

Google Mail: Made it easy to stay in touch with team members and external partners.

AI Tools (Google Gemini, GPT-O1, Claude): To help troubleshoot code, optimise scripts, and suggest best practices.

GitHub: Code repository for version control and documentation

Roboflow: Combined smaller open-source datasets to collate a large, diverse 300,000 image dataset, saving time with pre-labeled images.